End-of-flight results: candidate graphics in Journal of Navigation form

Proofs: each figure at 127 mm width with its caption beneath. Greyscale, no title or notes on the image.

Davey model with the fuel model, final end-of-flight run of 7 November 2026. Booklet built 9 November 2026.

Part 1. Impact position, energy and new concepts

eofn-01-overlay-searched: Overlay of two treatments on one map, 50 % and 95 % regions, with the previously searched area. Develops attachment 2 with full latitude range and an inset.

eofn-02-overlay: As eofn-01 without the previously searched area.

eofn-03-overlay-searched-r600: As eofn-01, adds the 00:19:29 BFO only, as observed (dotted), which replicates the 18:25 log-on case.

eofn-04-overlay-50-99: As eofn-02 with 50 % and 99 % regions.

eofn-05-multiples-searched: Small multiples: three treatments in columns, whole region above and the high-probability box below, with the previously searched area. Develops attachment 3.

eofn-06-multiples-50-90-99: As eofn-05 with 50/90/99 % regions and no search overlay.

eofn-07-multiples-two: Two treatments only (BTO only; Holland transient bias added), 2 × 2 panels, with the previously searched area.

eofn-08-energy-dots: Impact kinetic energy classes as dots over the 50/95 % contours, two treatments, with cumulative distributions of energy and peak impact power. Develops attachment 6.

eofn-09-power-dots: As eofn-08 with peak impact power classes. Develops attachment 5.

eofn-10-energy-share: Share of impacts at 4 GJ or more mapped as a field, with share by probability region.

eofn-11-map-marginals: New concept: one map with the latitude distribution (b) on the right and the distance from the 7th arc (c) as an inset.

eofn-12-corner: New concept: corner plot of impact latitude, distance from the 7th arc, time to impact and kinetic energy.

Part 2. 00:19 descent-rate chart (attachment 1)

descent-0019-A1-evidence-dots: Right-hand space: ln B for each treatment with both BFOs, both route models, against the weak-evidence band.

descent-0019-A2-prediction-contours: Right-hand space: posterior probabilities with an equal prior; contours show the descent rates predicted from the BTOs only.

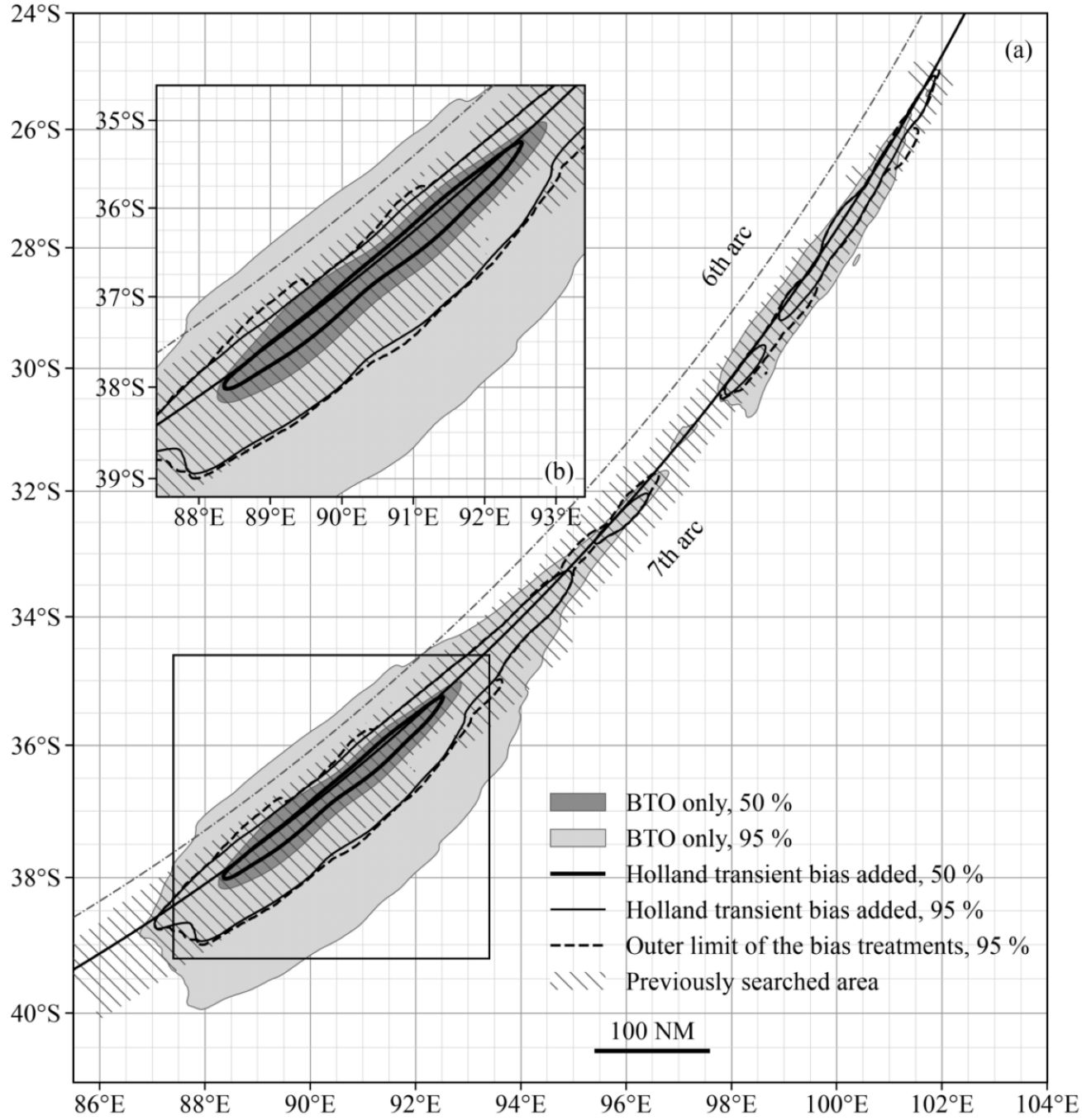
descent-0019-A3-scorecard: Right-hand space: test scorecard (evidence, 18:25 log-on fit, accident-dive areas, Boeing build-ups). Sub-agent recommendation for Part A.

descent-0019-A4-compact-table: Right-hand space: compact numeric table (ln B, posterior probability, effective samples).

descent-0019-B1-slope-bands: Whole-graphic alternative: descent-rate change over 8 s from the implied 00:19:29 rate.

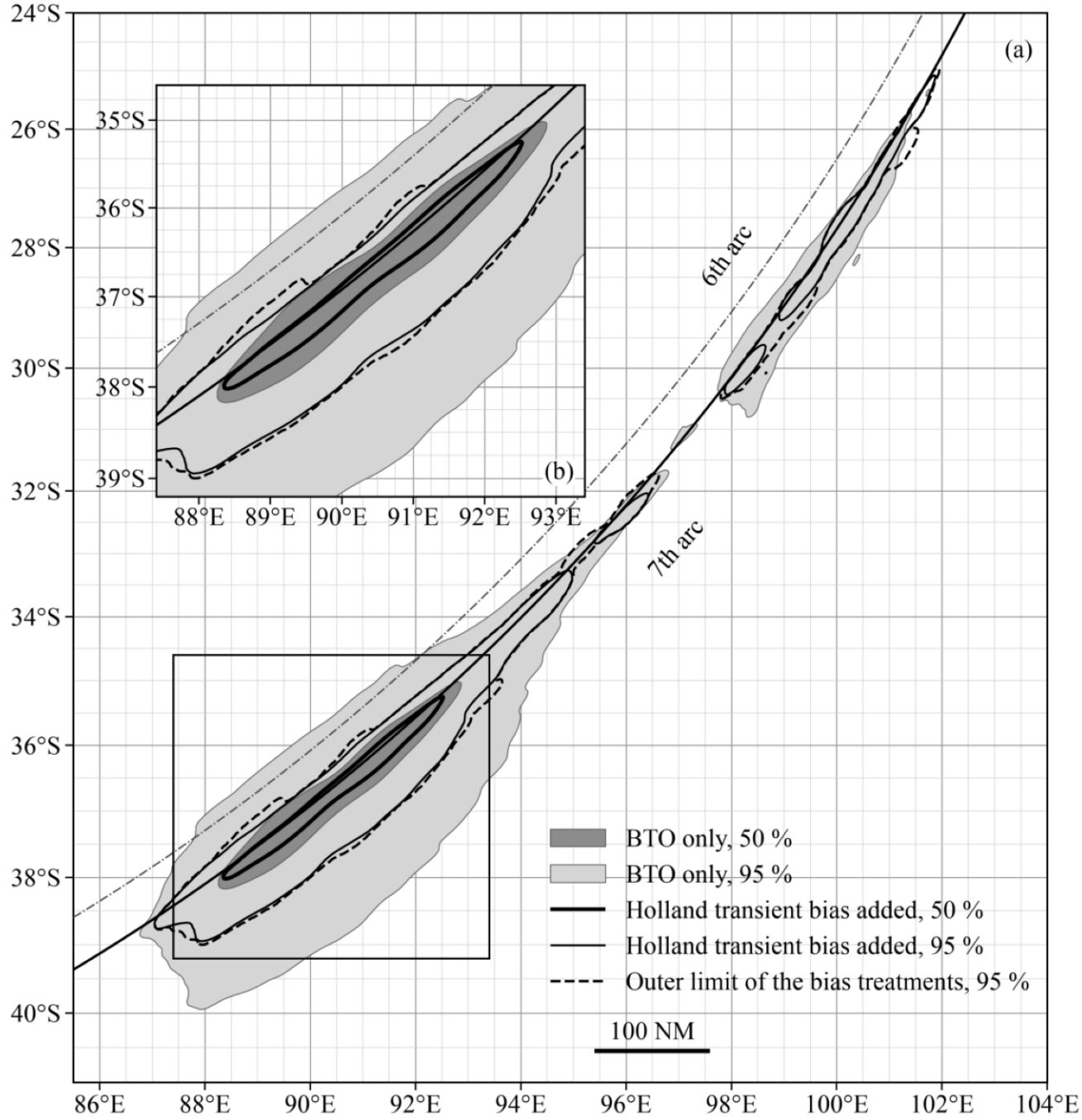
descent-0019-B2-acceleration-plane: Whole-graphic alternative: descent rate against mean downward acceleration (g).

descent-0019-B3-conditional-cut: Whole-graphic alternative: dot plot per accident case and per Boeing simulation, at the two implied 00:19:29 rates. Sub-agent recommendation for Part B.

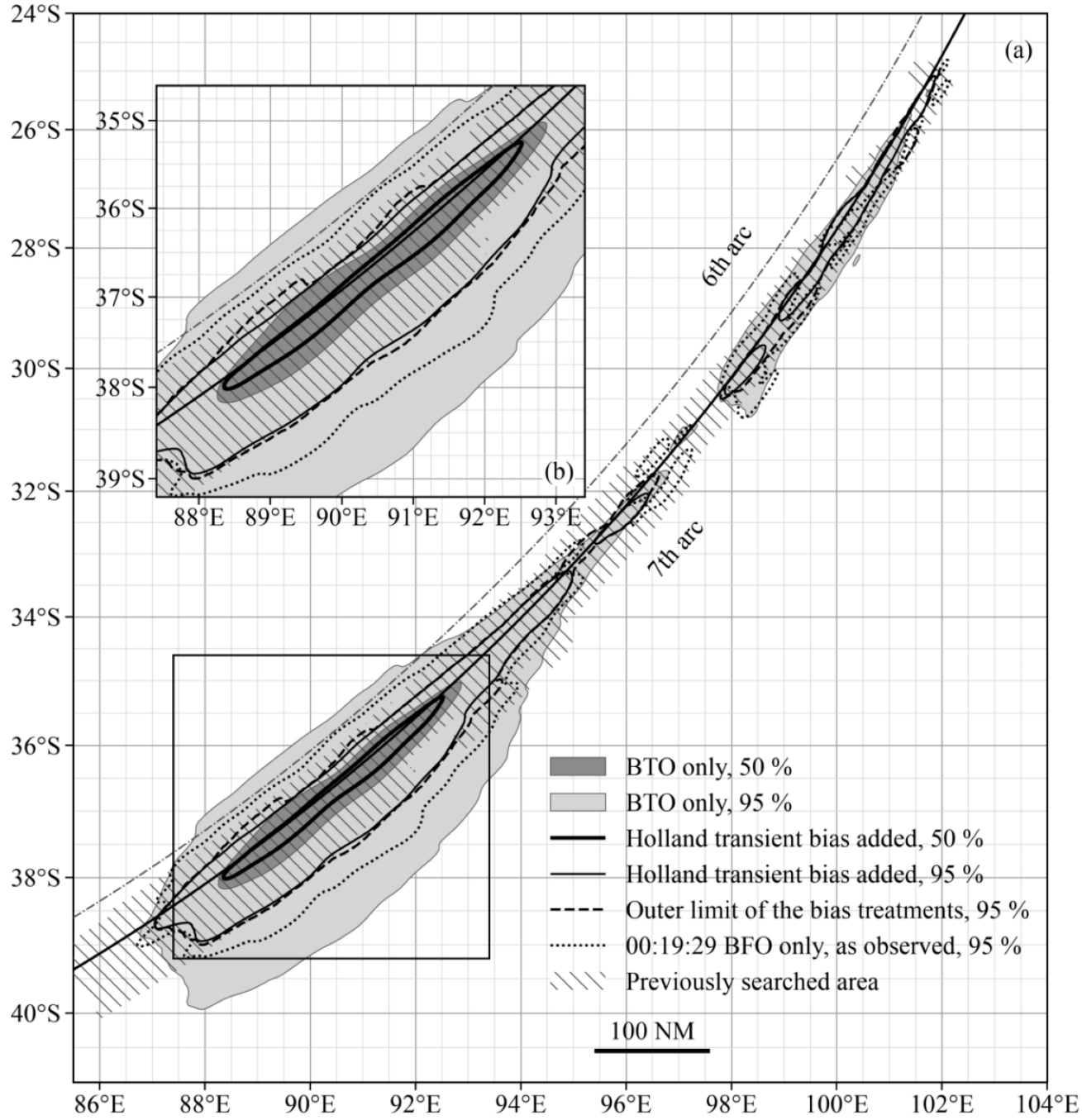


Impact position x_{Impact}, Davey model with the fuel model, for two treatments of the 00:19 satellite data: (a) whole region; (b) the area of highest probability (box in a). Light and dark grey: smallest regions holding 95 % and 50 % of the probability with the BTO values only. Thin and heavy black lines: the same regions with both BFOs and Holland's transient bias added. Dashed line: outer limit of the 95 % regions of all treatments that add a start-up bias to the BFO. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft. Hatching: previously searched area (ATSB Phase 2, Ocean Infinity 2018 and 2025–26). Mercator projection.

eofn-02-overlay

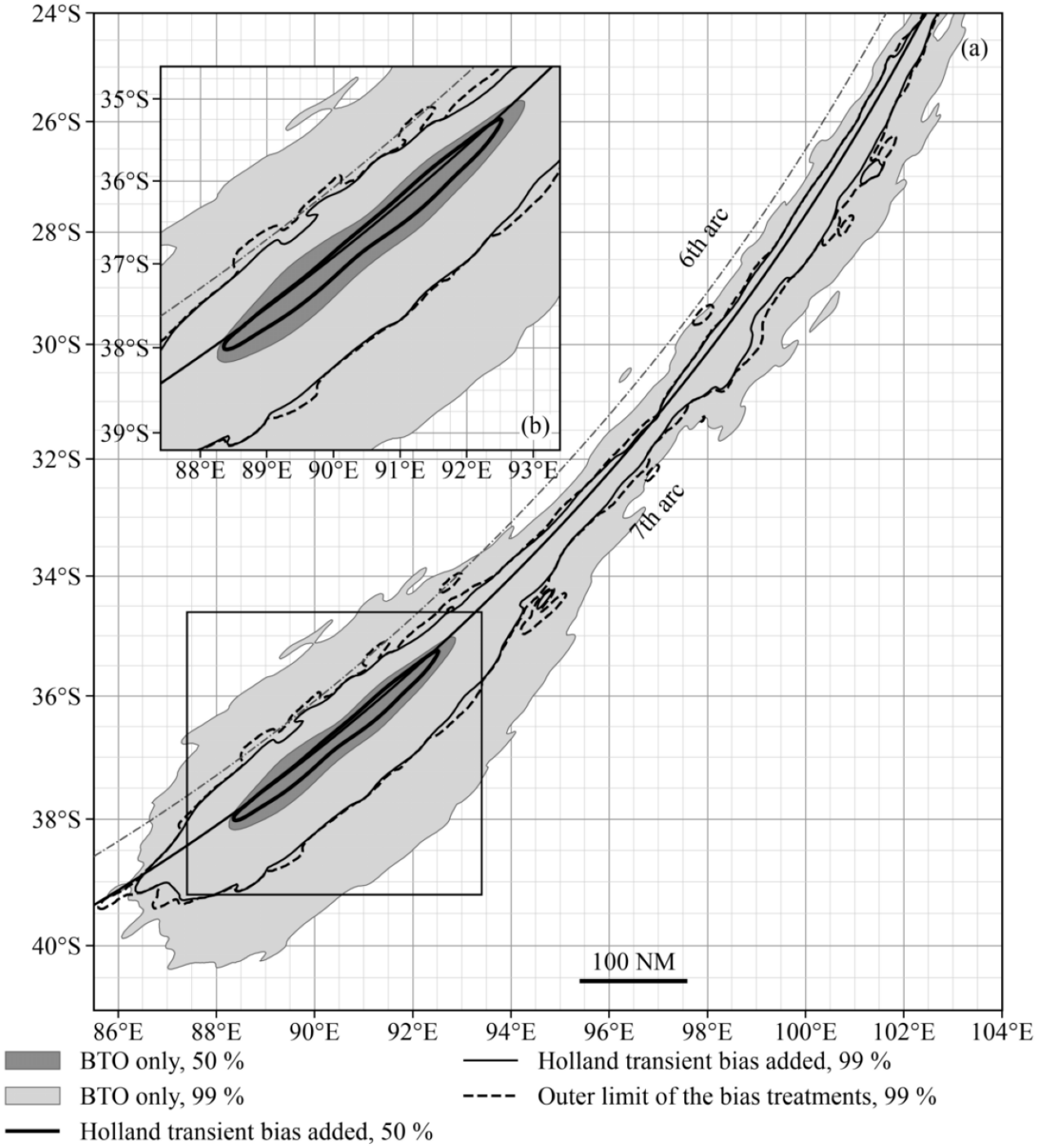


Impact position x_{Impact}, Davey model with the fuel model, for two treatments of the 00:19 satellite data: (a) whole region; (b) the area of highest probability (box in a). Light and dark grey: smallest regions holding 95 % and 50 % of the probability with the BTO values only. Thin and heavy black lines: the same regions with both BFOs and Holland's transient bias added. Dashed line: outer limit of the 95 % regions of all treatments that add a start-up bias to the BFO. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft. Mercator projection.

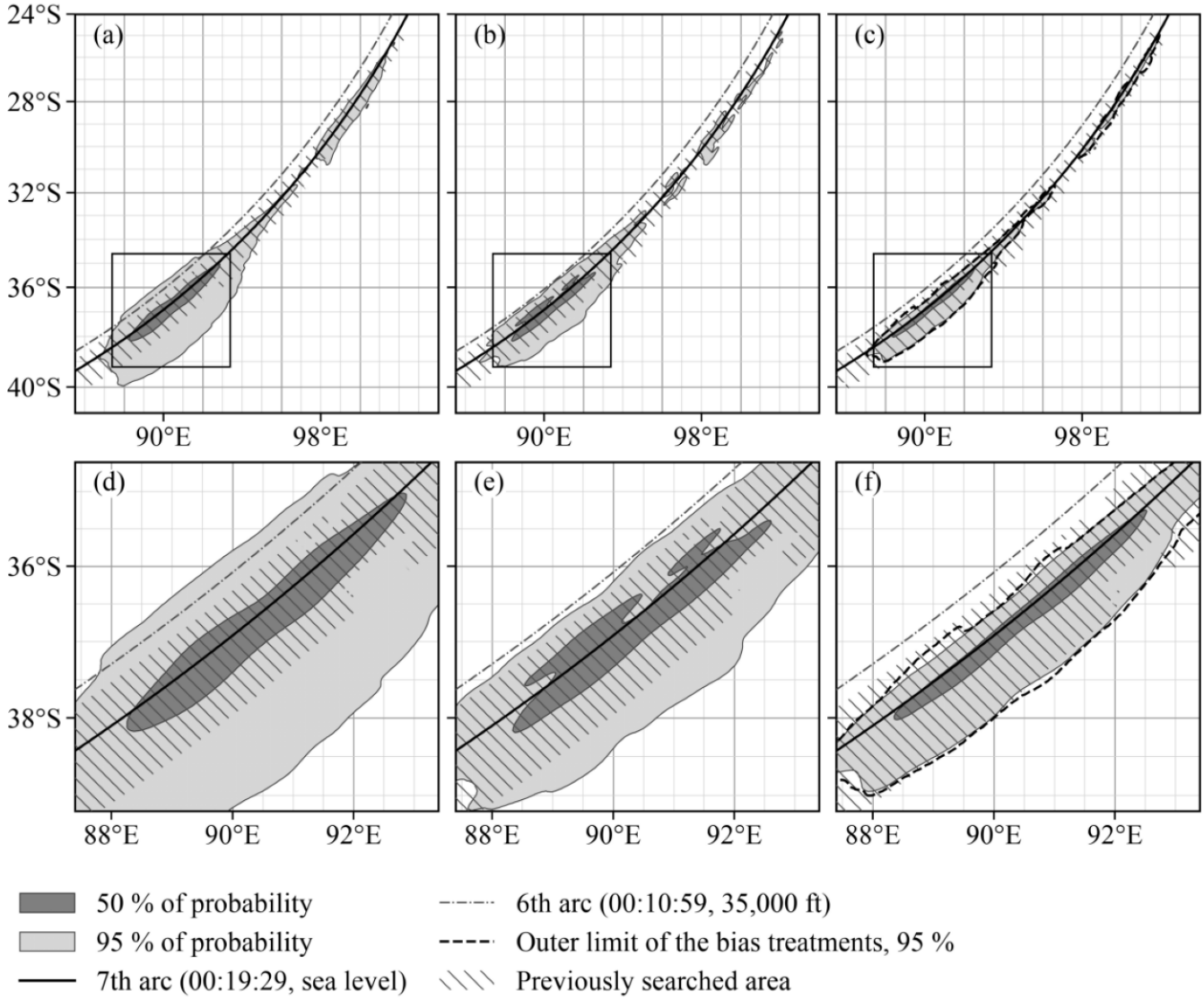


Impact position x_{Impact}, Davey model with the fuel model, for two treatments of the 00:19 satellite data: (a) whole region; (b) the area of highest probability (box in a). Light and dark grey: smallest regions holding 95 % and 50 % of the probability with the BTO values only. Thin and heavy black lines: the same regions with both BFOs and Holland's transient bias added. Dashed line: outer limit of the 95 % regions of all treatments that add a start-up bias to the BFO. Dotted line: the 95 % region with the 00:19:29 BFO only, as observed. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft. Hatching: previously searched area (ATSB Phase 2, Ocean Infinity 2018 and 2025–26). Mercator projection.

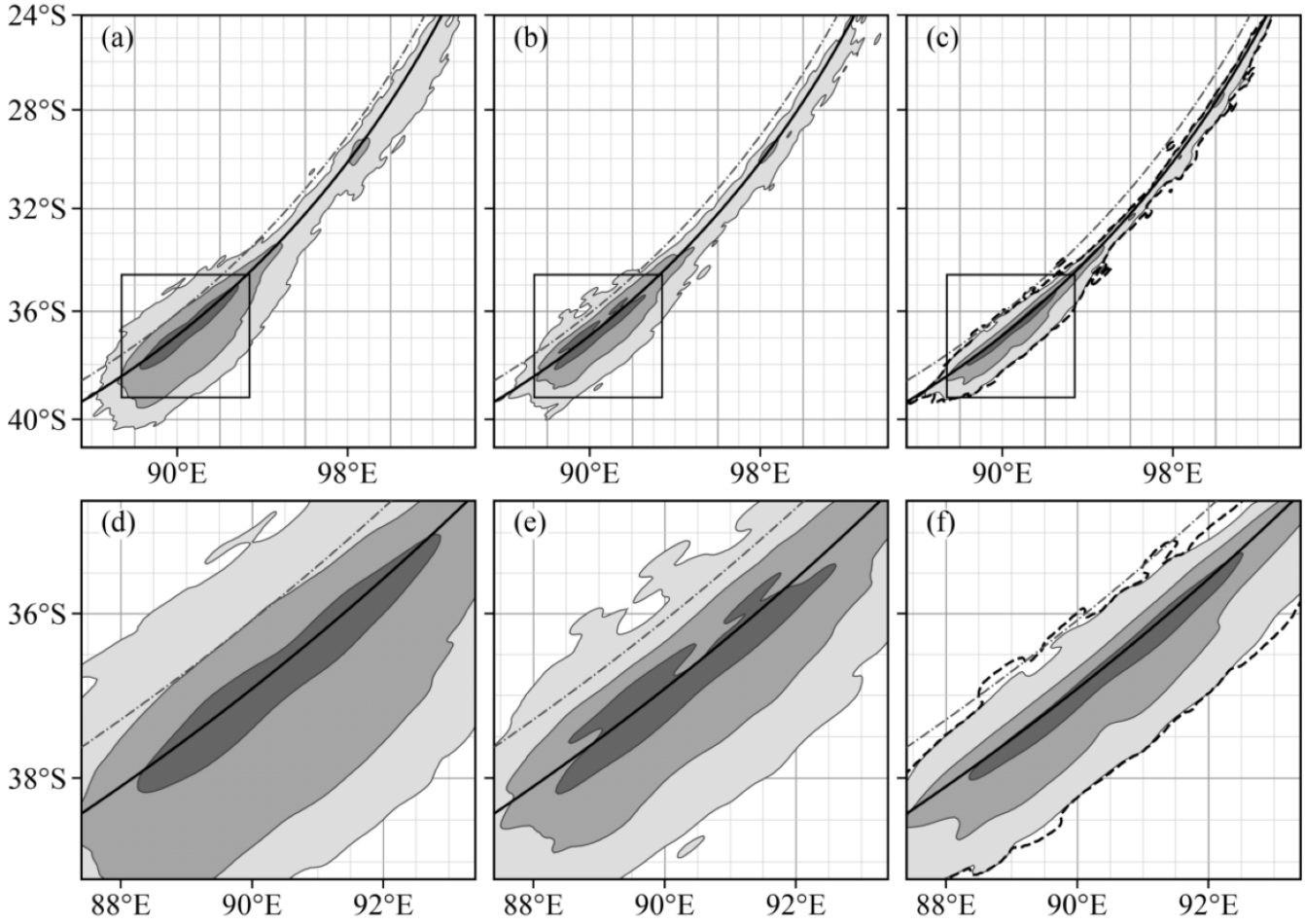
eofn-04-overlay-50-99



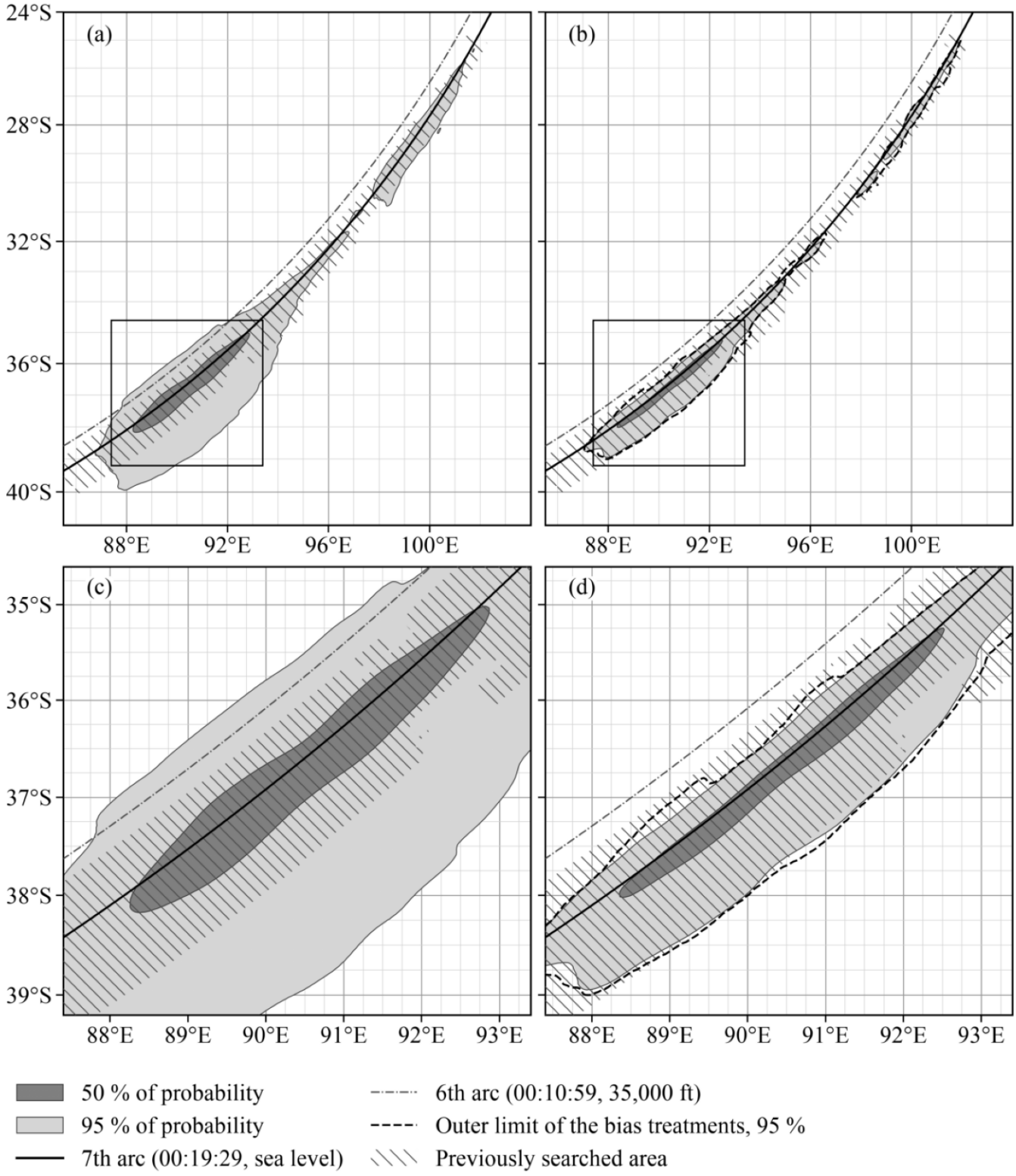
Impact position x_{Impact}, Davey model with the fuel model, for two treatments of the 00:19 satellite data: (a) whole region; (b) the area of highest probability (box in a). Light and dark grey: smallest regions holding 99 % and 50 % of the probability with the BTO values only. Thin and heavy black lines: the same regions with both BFOs and Holland's transient bias added. Dashed line: outer limit of the 99 % regions of all treatments that add a start-up bias to the BFO. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft. Mercator projection.



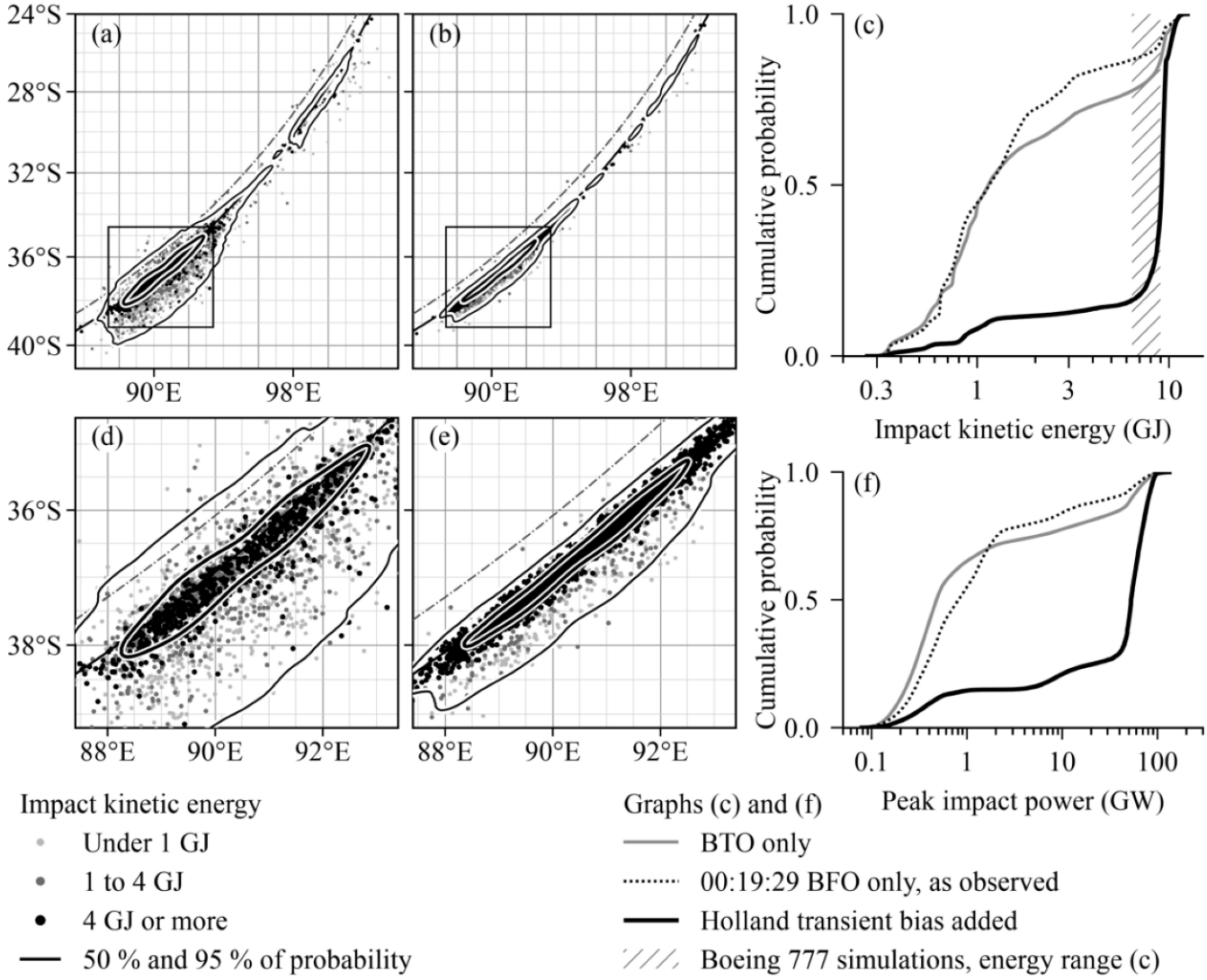
Impact position x_{Impact}, Davey model with the fuel model, for three treatments of the 00:19 satellite data: (a, d) BTO values only; (b, e) BTO values and the 00:19:29 BFO as observed; (c, f) BTO values and both BFOs with Holland's transient bias added. The upper row shows the whole region; the lower row shows the box marked above. Shading: smallest regions holding 50 % and 95 % of the probability. Dashed line in (c, f): outer limit of the 95 % regions of all treatments that add a start-up bias. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft. Hatching: previously searched area (ATSB Phase 2, Ocean Infinity 2018 and 2025–26).



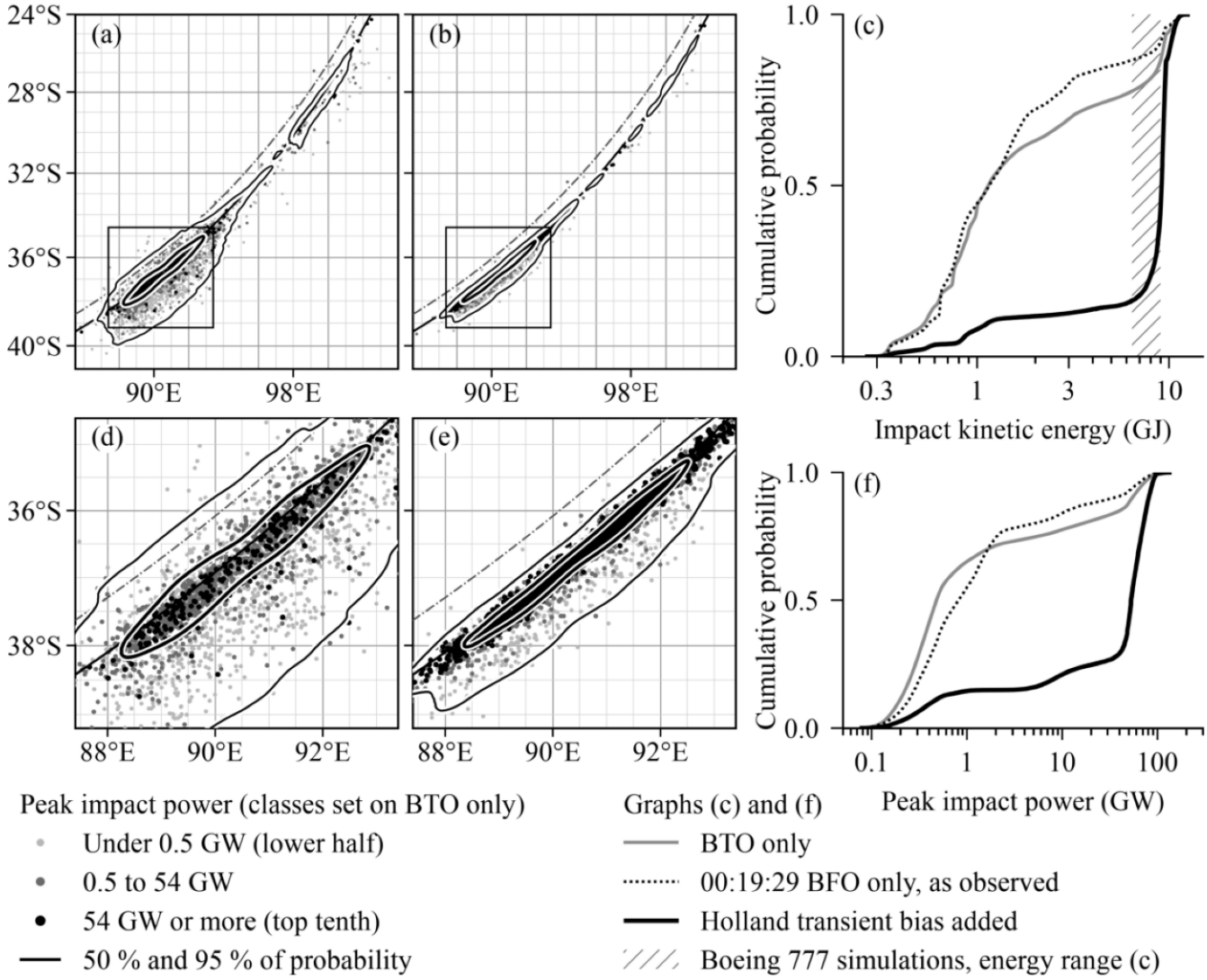
Impact position x_{Impact}, Davey model with the fuel model, for three treatments of the 00:19 satellite data: (a, d) BTO values only; (b, e) BTO values and the 00:19:29 BFO as observed; (c, f) BTO values and both BFOs with Holland's transient bias added. The upper row shows the whole region; the lower row shows the box marked above. Shading: smallest regions holding 50 %, 90 % and 99 % of the probability. Dashed line in (c, f): outer limit of the 99 % regions of all treatments that add a start-up bias. Solid curve: 7th arc at sea level; dash-dot curve: 6th arc at 35,000 ft.



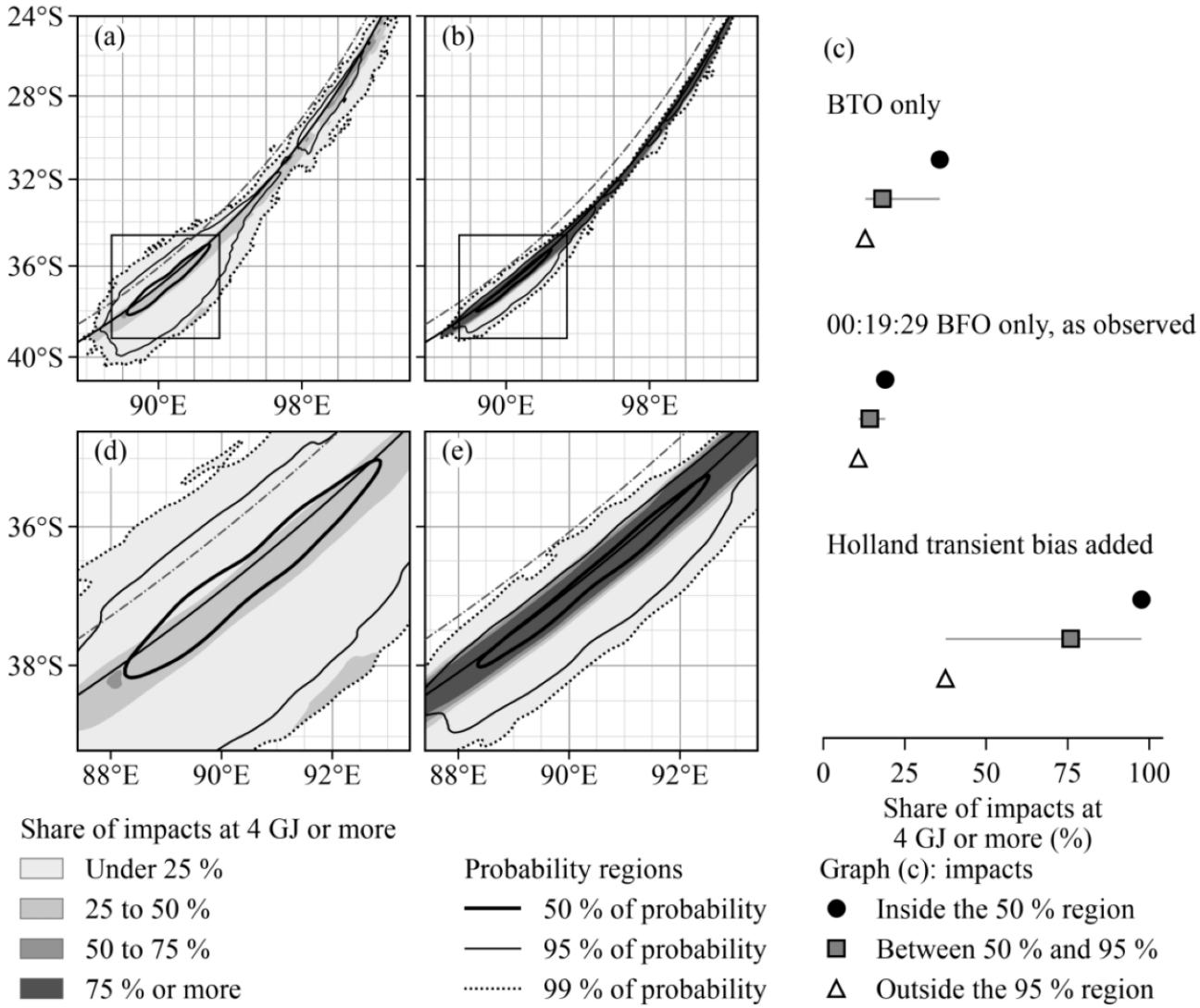
Impact position x_{Impact}, Davey model with the fuel model: (a, c) BTO values only; (b, d) BTO values and both BFOs with Holland's transient bias added. The upper row shows the whole region; the lower row shows the box marked above. Shading: smallest regions holding 50 % and 95 % of the probability. Dashed line in (b, d): outer limit of the 95 % regions of all treatments that add a start-up bias. Solid curve: 7th arc; dash-dot curve: 6th arc. Hatching: previously searched area (ATSB Phase 2, Ocean Infinity 2018 and 2025–26).



Simulated impacts by kinetic energy, Davey model with the fuel model: (a, d) BTO values only; (b, e) BTO values and both BFOs with Holland's transient bias added; (d, e) show the box marked in (a, b). Each dot is one of 3,000 impacts drawn in proportion to probability: light grey under 1 GJ, mid-grey 1 to 4 GJ, black 4 GJ or more. Black lines with a white edge enclose 50 % and 95 % of the probability. (c) Cumulative distribution of impact kinetic energy; the hatched band is the range of the Boeing 777 simulations. (f) Cumulative distribution of peak impact power. In (c, f): grey line, BTO only; dotted line, 00:19:29 BFO only, as observed; heavy line, Holland's bias.

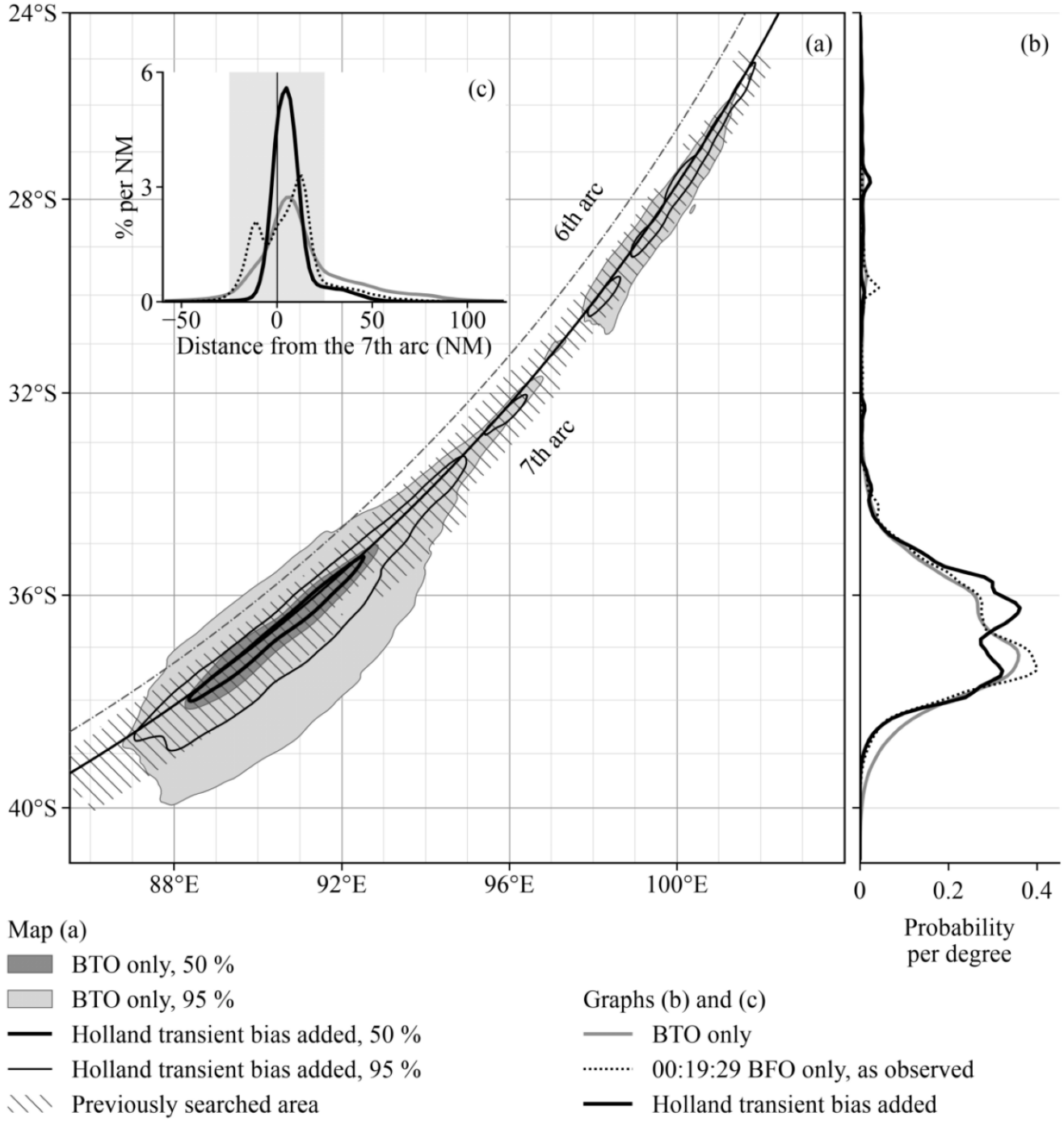


Simulated impacts by peak impact power, Davey model with the fuel model: (a, d) BTO values only; (b, e) BTO values and both BFOs with Holland's transient bias added; (d, e) show the box marked in (a, b). Each dot is one of 3,000 impacts drawn in proportion to probability. Classes are set on the BTO-only distribution: light grey, lower half; mid-grey, 50th to 90th percentile; black, top tenth. Black lines with a white edge enclose 50 % and 95 % of the probability. (c) and (f) as in the energy figure.

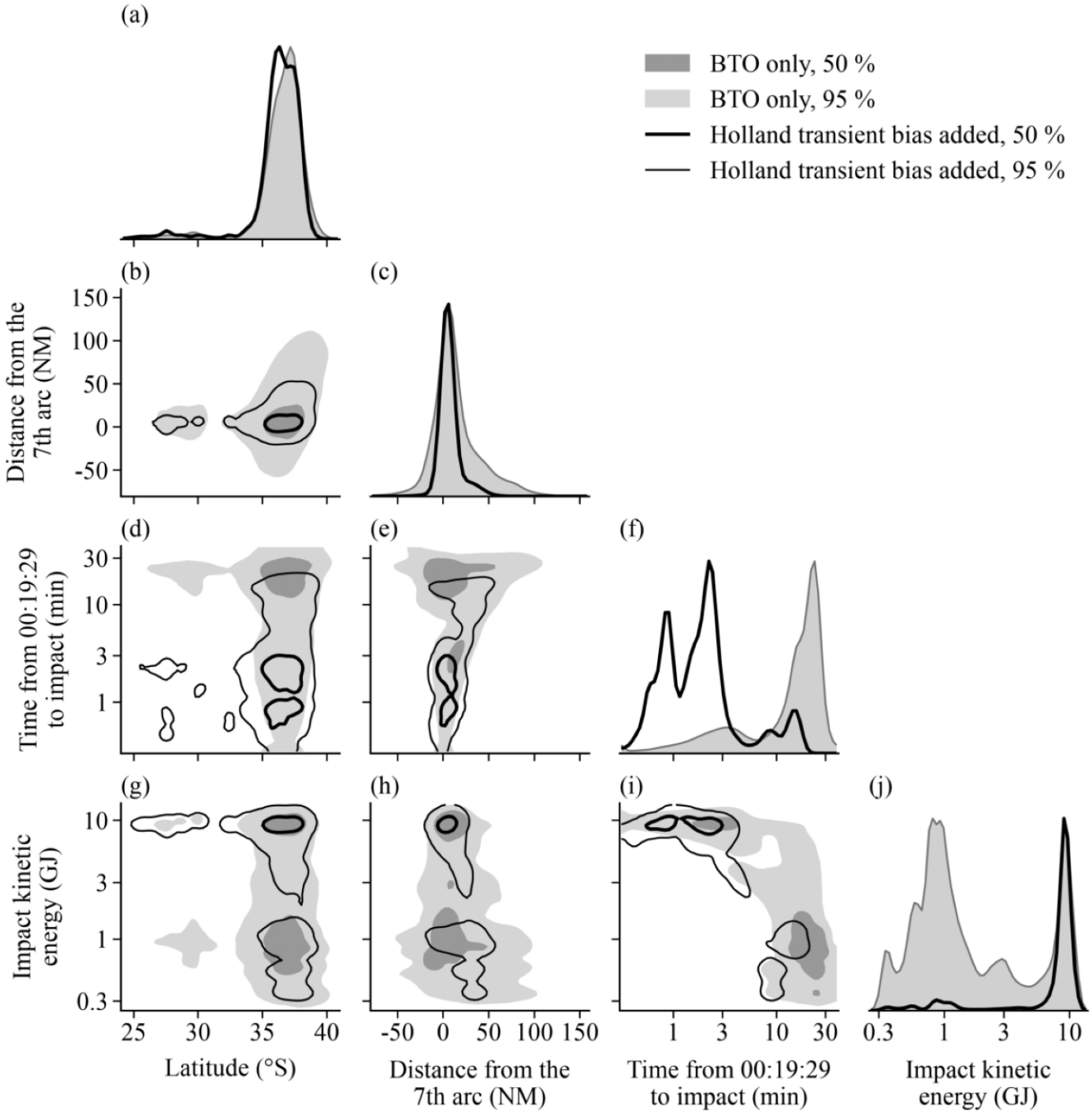


Where the high-energy impacts lie, Davey model with the fuel model: (a, d) BTO values only; (b, e) BTO values and both BFOs with Holland's transient bias added; (d, e) show the box marked in (a, b). Shading gives the share of impacts with a kinetic energy of 4 GJ or more at each place, inside the 99 % probability region. Heavy, thin and dotted lines enclose 50 %, 95 % and 99 % of the probability. (c) Share of impacts at 4 GJ or more inside the 50 % region (circles), between 50 % and 95 % (squares) and outside the 95 % region (triangles); the grey line joins the three shares of each treatment.

eofn-11-map-marginals

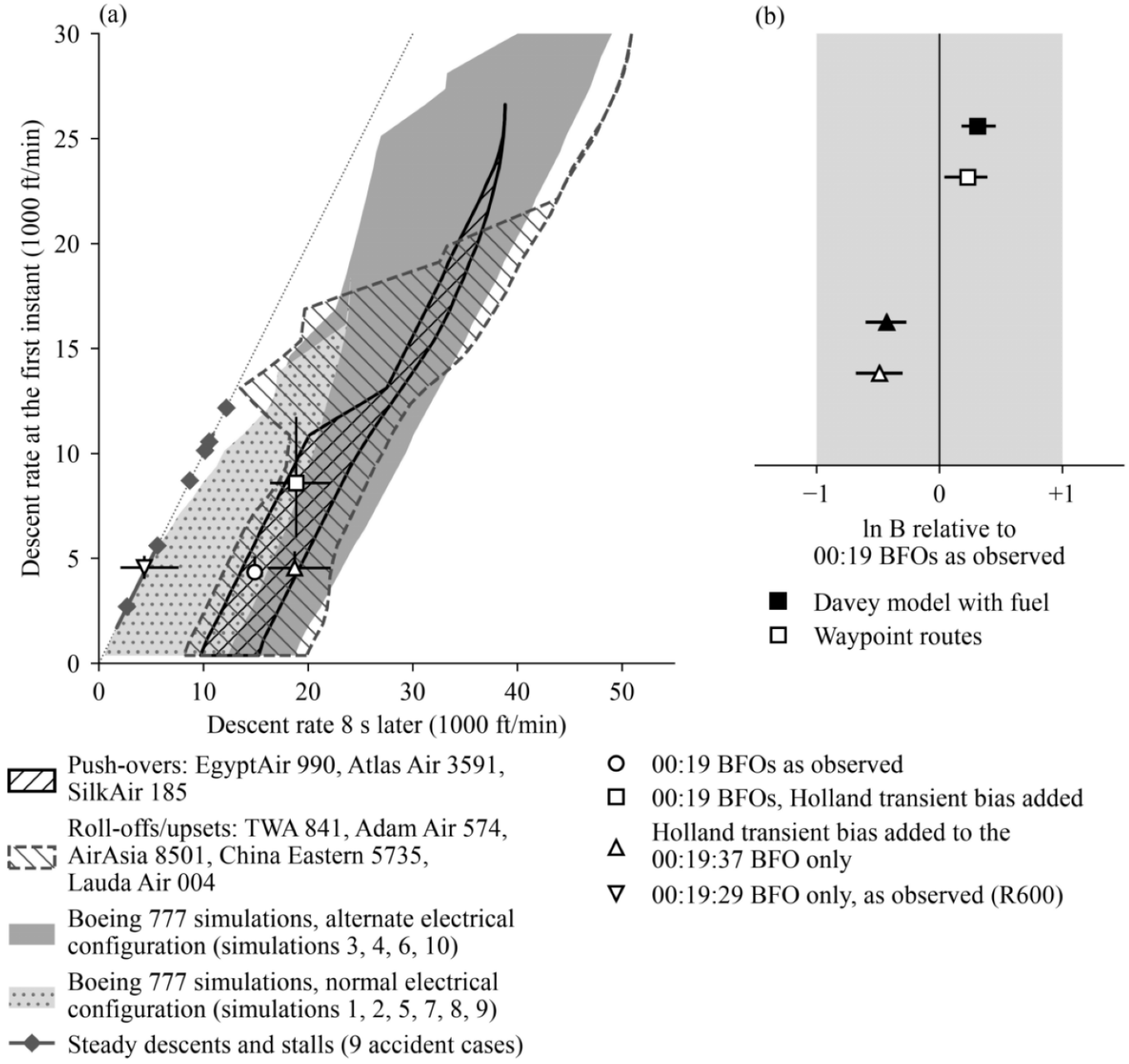


Impact position x_{Impact} and its distributions, Davey model with the fuel model. (a) Map: light and dark grey, regions holding 95 % and 50 % of the probability with the BTO values only; thin and heavy lines, the same regions with both BFOs and Holland's transient bias added; solid curve, 7th arc; dash-dot curve, 6th arc. Hatching: previously searched area (ATSB Phase 2, Ocean Infinity 2018 and 2025–26). (b) Probability per degree of latitude, on the latitude scale of (a). (c) Probability per nautical mile of distance from the 7th arc (positive to the south-east); the shaded band is within 25 NM. In (b) and (c): grey line, BTO only; dotted line, 00:19:29 BFO only, as observed; heavy line, Holland's bias.



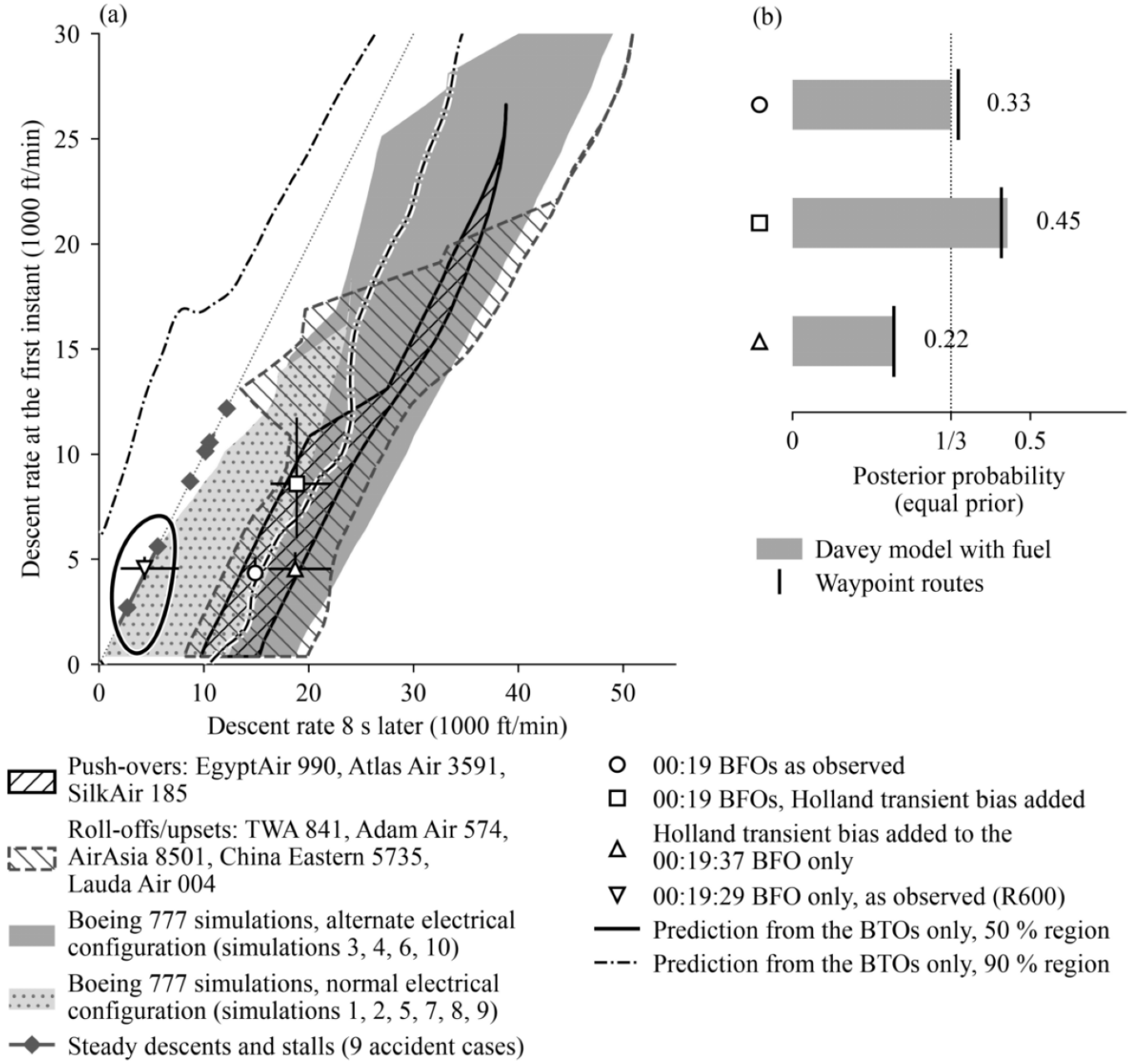
Joint distributions of four impact quantities, Davey model with the fuel model: latitude, distance from the 7th arc (positive to the south-east), time from 00:19:29 to impact, and impact kinetic energy. Panels (a), (c), (f) and (j) show each quantity alone, scaled to its peak; the other panels show pairs. Grey shading: regions holding 95 % and 50 % of the probability with the BTO values only. Black lines: the same regions with both BFOs and Holland's transient bias added. Time and energy use logarithmic scales.

descent-0019-A1-evidence-dots

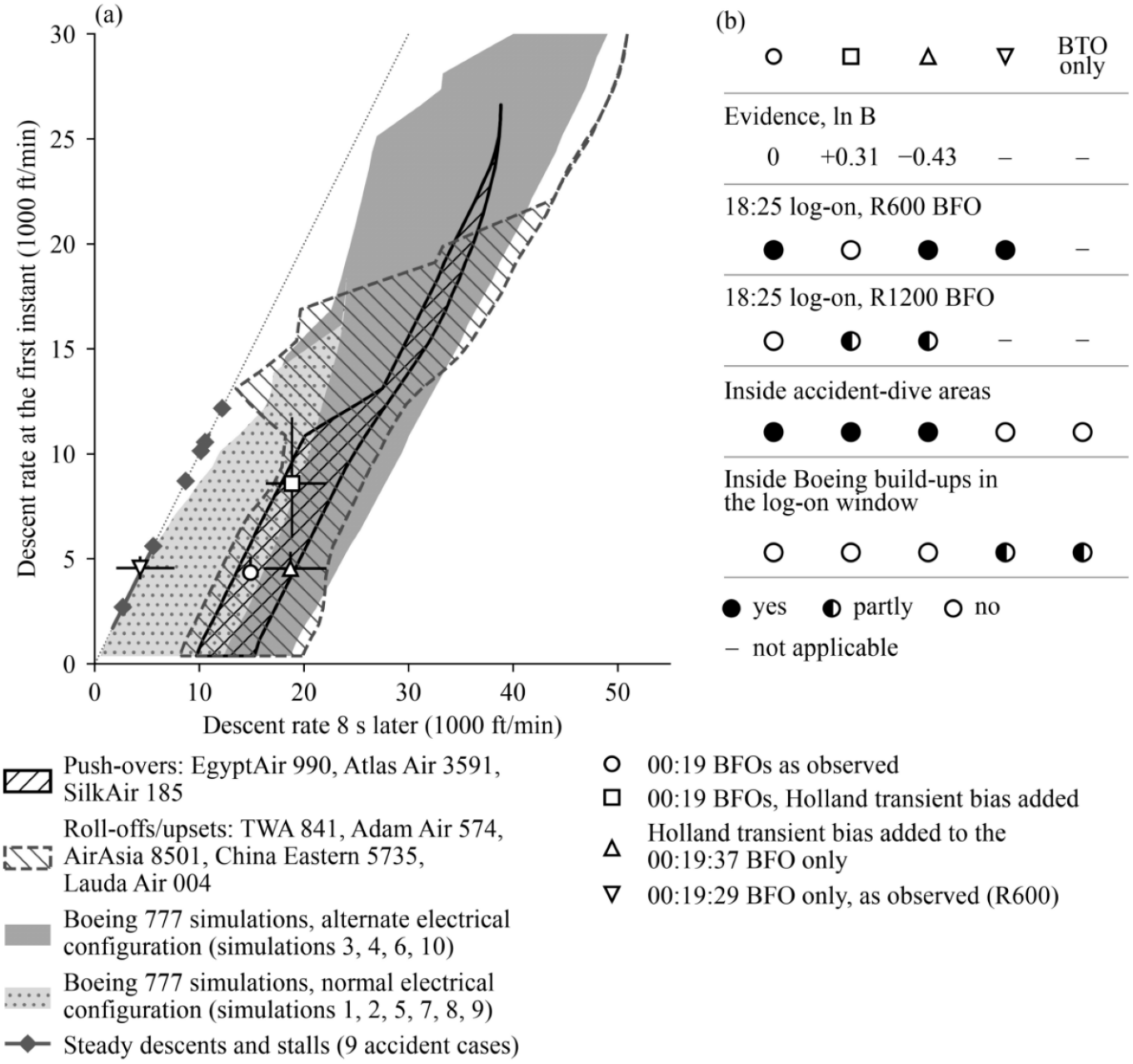


(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) Log Bayes factor against the 00:19 BFOs as observed, for the same data; lines are 90 % intervals over seeds. The grey band ($|\ln B| < 1$) is evidence not worth more than a bare mention (Kass and Raftery, 1995). R600 is the 600 bit/s channel of the 00:19:29 burst.

descent-0019-A2-prediction-contours

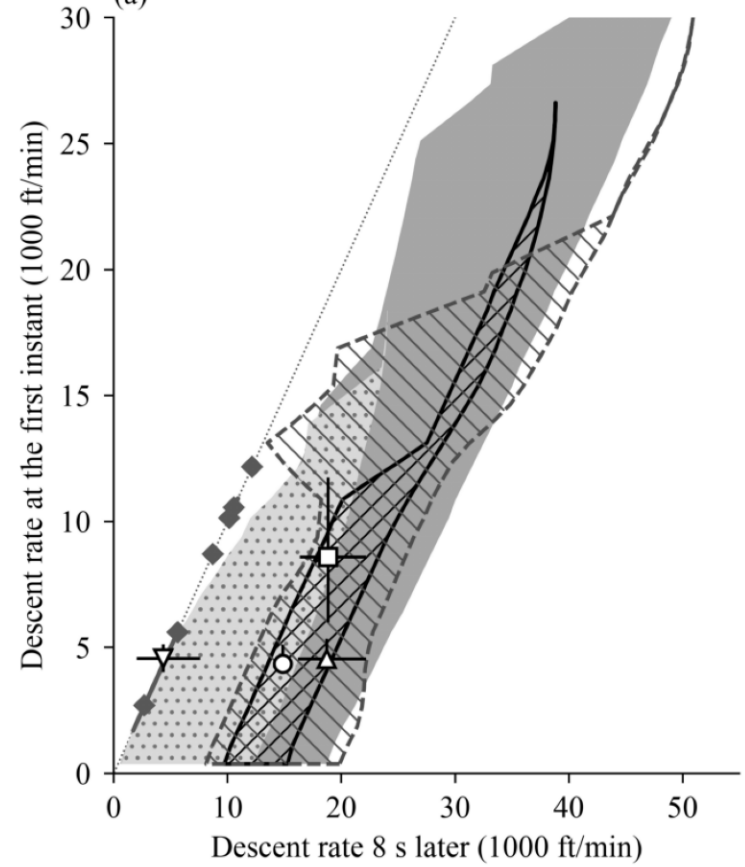


(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. Contours enclose 50 % and 90 % of the rates predicted from the 00:19 BTOs alone for descending paths (27 % of the predicted weight). (b) Posterior probability, with an equal prior (dotted line), of the three treatments that use both BFOs. R600 is the 600 bit/s channel of the 00:19:29 burst.



(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) Tests of each treatment: ln B against the BFOs as observed (only for treatments with both BFOs); fit when applied to the 18:25 log-on (half: fit only at the top of Holland's range, which was set from that log-on); posterior weight inside the areas (filled ≥ 90 %, half 10–90 %, open < 10 %). R600 and R1200 are the 600 and 1200 bit/s channels.

(a)



Push-overs: EgyptAir 990, Atlas Air 3591, SilkAir 185

Roll-offs/upsets: TWA 841, Adam Air 574, AirAsia 8501, China Eastern 5735, Lauda Air 004

Boeing 777 simulations, alternate electrical configuration (simulations 3, 4, 6, 10)

Boeing 777 simulations, normal electrical configuration (simulations 1, 2, 5, 7, 8, 9)

Steady descents and stalls (9 accident cases)

○ 00:19 BFOs as observed

□ 00:19 BFOs, Holland transient bias added

△ Holland transient bias added to the 00:19:37 BFO only

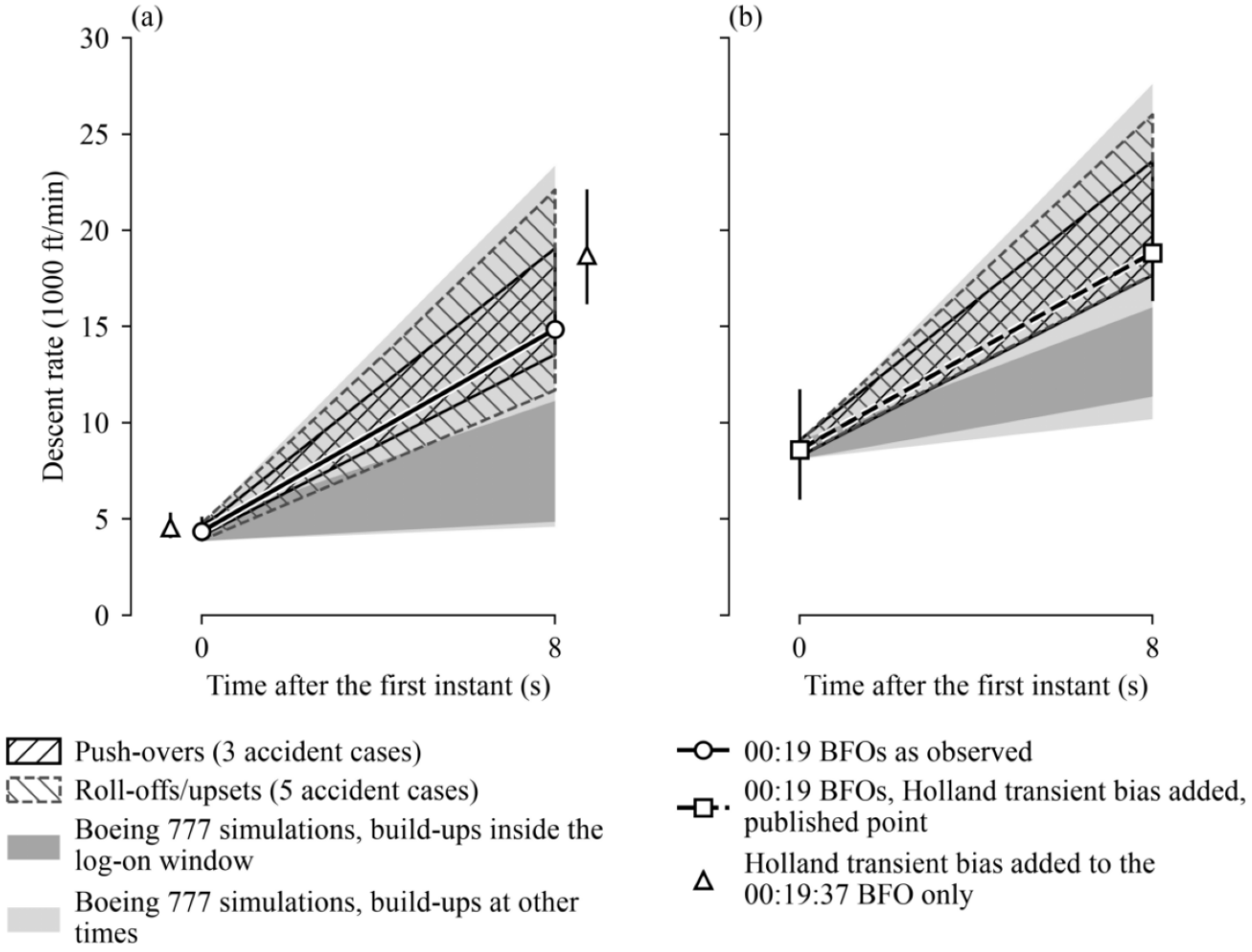
▽ 00:19:29 BFO only, as observed (R600)

(b)

	ln B (90 % interval)	Posterior probability	Effective samples
○	0 (reference)	0.33	508*
□	+0.31 (+0.18, +0.46)	0.45	3,347
△	−0.43 (−0.60, −0.27)	0.22	1,357*
▽	–	–	22,056
BTO only	–	–	499,760

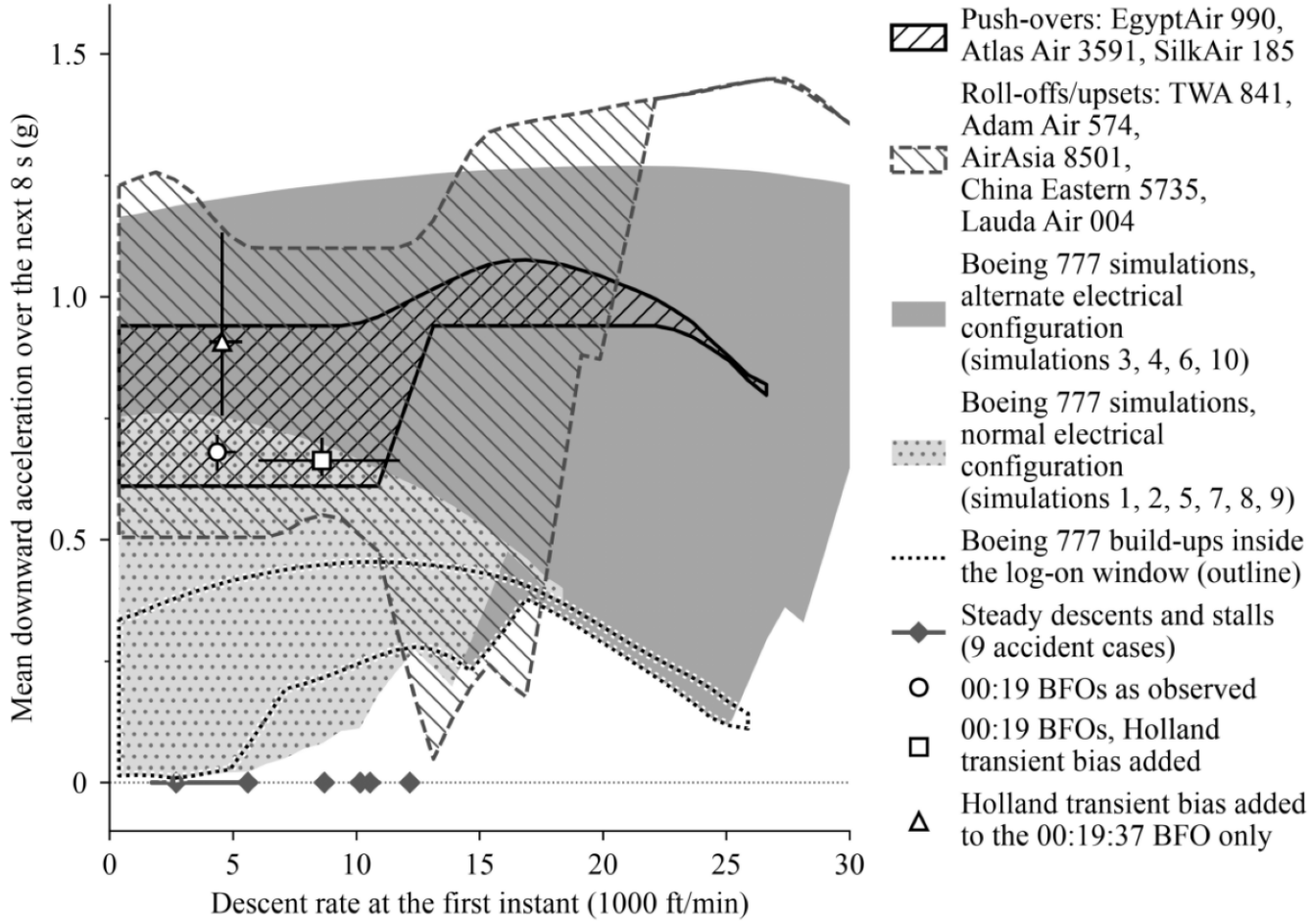
(a) Descent rate at a first instant and 8 s later during dive build-ups; each area covers one group of dives, and the dotted line is a steady descent. Symbols mark the 00:19:29 and 00:19:37 rates implied by each 00:19 BFO treatment; bars are 5–95 % posterior ranges (Davey model with fuel), centred on the published point for Holland transient bias. (b) ln B against the BFOs as observed (90 % interval over seeds) and posterior probability with an equal prior, both only for treatments with both BFOs; effective sample size of the final run (* fewer than 500 in the no-pilot family). R600 is the 600 bit/s channel of the 00:19:29 burst.

descent-0019-B1-slope-bands

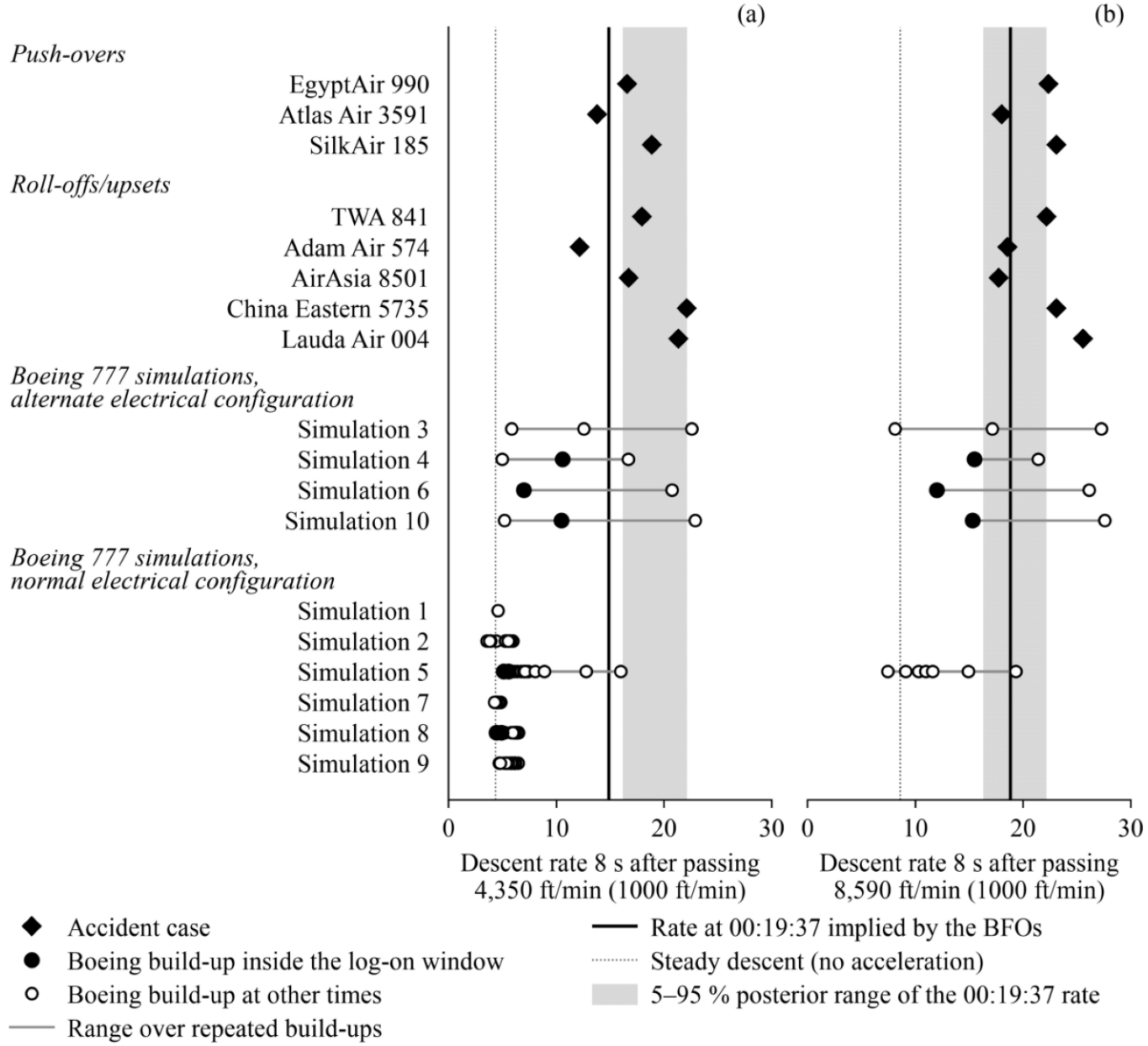


Change of descent rate over 8 s in every dive build-up whose first-instant rate lies within 500 ft/min of the 00:19:29 rate implied by (a) the 00:19 BFOs as observed (4,350 ft/min) and (b) the BFOs with Holland transient bias added (8,590 ft/min, published point). Each area spans all such changes in one group. Heavy lines join the implied rates; bars are 5–95 % posterior ranges at each instant (Davey model with fuel), and triangles give the bias added to the 00:19:37 BFO only. The log-on window is 67–198 s after power loss.

descent-0019-B2-acceleration-plane



Descent rate at a first instant and mean downward acceleration over the next 8 s (g = 32.174 ft/s²) during the build-up of dives; each area covers one group, and steady descents lie on the dotted zero line. The dotted outline encloses Boeing build-ups inside the log-on window (67–198 s after power loss). Symbols are the values implied by the 00:19 BFO treatments (0.68 g as observed; 0.66 g with Holland transient bias added, published point); bars give the 5–95 % posterior range (Davey model with fuel).



Descent rate 8 s after each dive passed the 00:19:29 rate implied by (a) the 00:19 BFOs as observed (4,350 ft/min) and (b) the BFOs with Holland transient bias added (8,590 ft/min, published point). Solid lines: implied 00:19:37 rates (14,860 and 18,830 ft/min). Grey bands: 5–95 % posterior range of the 00:19:37 rate with (a) the bias added to the 00:19:37 BFO only and (b) the bias added to both (Davey model with fuel). Filled circles are Boeing build-ups inside the log-on window (67–198 s after power loss); a row without a symbol never reaches that rate.